CS341 · COMPUTER GRAPHICS

Mock Midterm Solutions

20 marks · the same distribution as the exam

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01 Solution Q1: true / false

#	Answer	Reason
1	**True**	Raster stores pixel values in the frame (refresh) buffer.
2	**False**	Plasma is emissive; LCD is the nonemissive one.
3	**False**	To one-tenth, not one-half.
4	**True**	`v = Q − P`
5	**False**	The reverse: y increases by 1 and x by $\frac{1}{m}$.

02 Solution Q2: multiple choice

#	Answer	Reason
1	**b**	The others are input devices.
2	**b**	The dot pitch slide: less than 0.31 mm.
3	**c**	`cos 90° = 0`
4	**c**	
5	**b**	(xmin, ymin) lower-left and (xmax, ymax) upper-right.

03 Solution Q3: short answers

1. List

Keyboard, mouse, trackball, joystick, data glove, digitizer, scanner, touch panel, light pen, voice system (any four).

2. Reason

A raster screen cannot draw a perfect line: each calculated point is rounded to the nearest pixel position, so every line except horizontal and vertical ones looks stair-stepped.

3. Comparison: random and raster

	Random scan	Raster scan
Drawing	The beam draws the lines directly	Sweeps the screen line by line, top to bottom
Storage	Display list	Frame buffer
Resolution	Very high, smooth lines	Stair-step lines
Color and shading	Limited	Easy

4. Comparison: active and passive

Active-matrix	Passive-matrix
A transistor at each pixel	One transistor controls a whole row or column
Brighter and sharper	Cheaper
More complicated and more expensive	Needs less power

04 Solution Q4: DDA

$dx = 6 - 1 = 5 \qquad dy = 4 - 1 = 3$

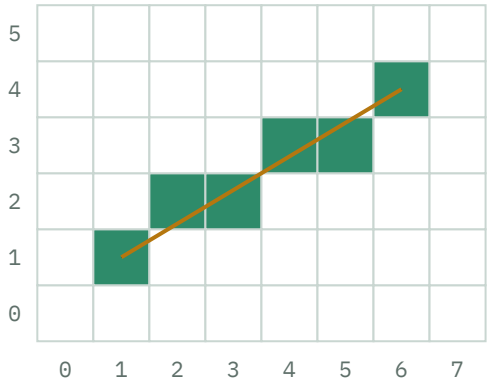
$m = \frac{3}{5} = 0.6 \rightarrow |m| < 1 \rightarrow \text{step in x}$

$steps = max(|dx|, |dy|) = 5$

$xInc = \frac{5}{5} = 1 \qquad yInc = \frac{3}{5} = 0.6$

$round(a) = int(a + 0.5)$

k	x	y	round(x)	round(y)
0	1	1	1	1
1	2	1.6	2	2
2	3	2.2	3	2
3	4	2.8	4	3
4	5	3.4	5	3
5	6	4	6	4



Points: (1,1) (2,2) (3,2) (4,3) (5,3) (6,4)

05 Solution Q5: aspect ratio

$$AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$$

$$= \frac{\frac{768}{6}}{\frac{1024}{8}}$$

$$= \frac{128}{128}$$

$$= 1$$

Vertical (Y) is always on top: vertical points ÷ height, over horizontal points ÷ width.